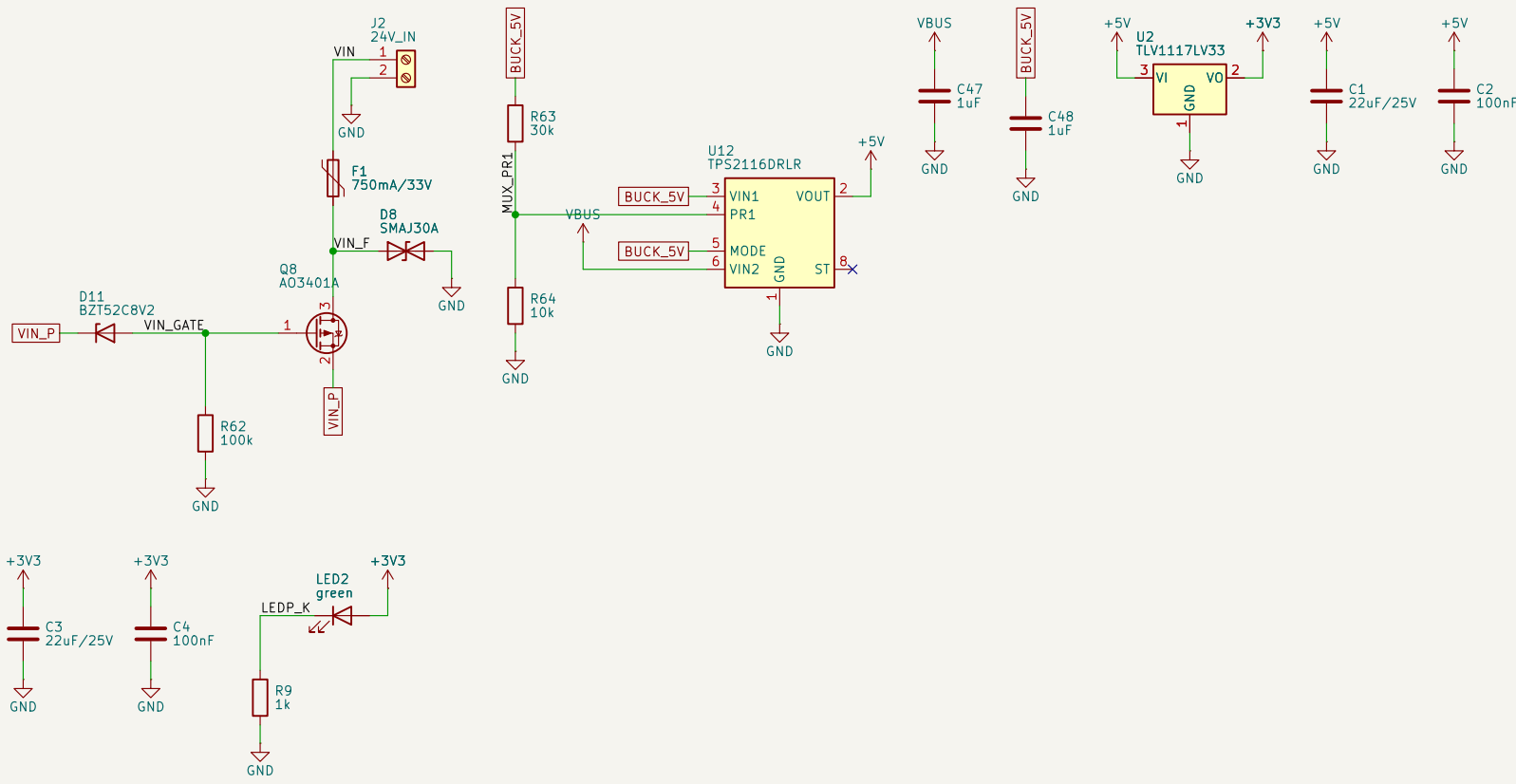
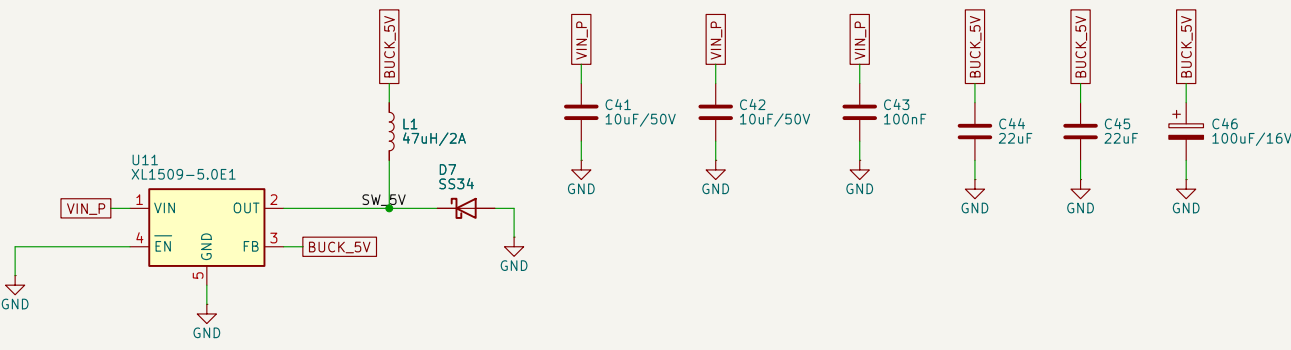


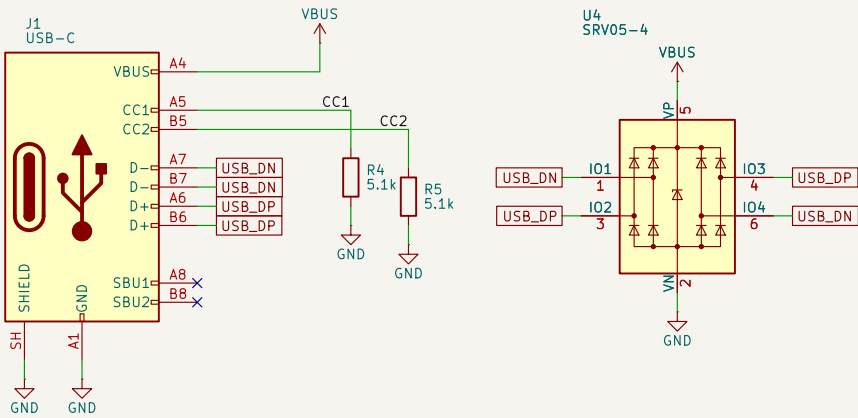
POWER IN 24 VDC terminal or USB  
F1+D8 clamp-and-blow at the entry. Q8 blocks reverse polarity as a P-FET, not a diode: its body diode conducts on correct polarity and the gate then pulls itself on through R62, clamped by D11 because 24 V is twice the A03401A's V<sub>GS</sub> rating. U11 (right) makes BUCK\_5V; U12 selects between it and USB VBUS to make +5V, giving the buck PRIORITY via R63/R64 rather than letting the higher voltage win -- a 5.25 V host would otherwise source the SSR terminals and the relay bank. U12 also blocks USB back-feeding the unpowered buck. C47/C48 are its input caps, one per channel, at the pins. U2 drops +5V to +3V3



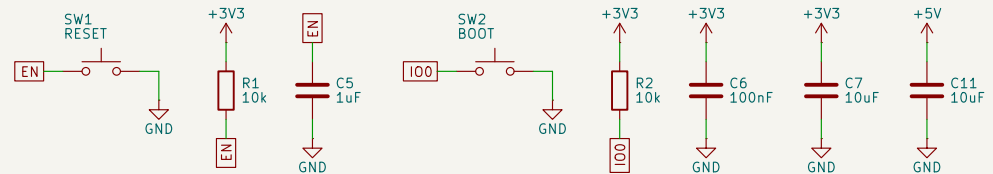
24 V -> 5 V BUCK: XL1509-5.0, fixed output  
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.



#### USB-C native USB flashing + ESD



#### RESET / BOOT / DECOUPLING



#### ESP32-S3-WROOM-1U-N16R2 GPIOs = firmware Kconfig defaults

